I built a test to catch AI cheating.

It caught me instead.

Some numbers can be proven wrong. Most can't — and that's the whole game.

KAGGLE / GOOGLE DEEPMIND 2026 SUBMISSION

Circuit-Theoretic Cognitive Bypass Detection

Does the declared reasoning actually produce the answer?

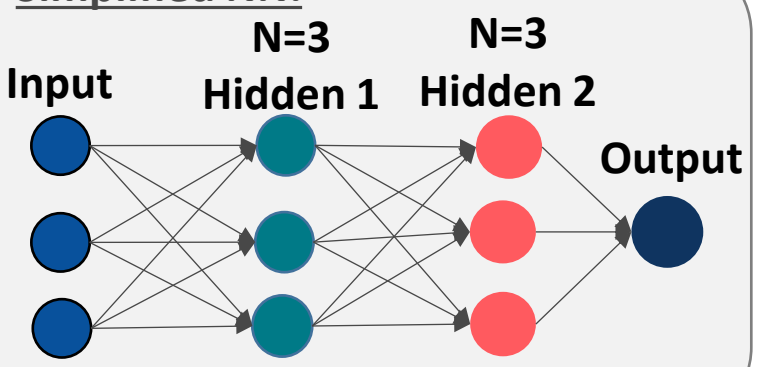
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THREE FIELDS OF STUDY · ONE PROBLEM

Gen AI

OpenAI's own researchers cannot explain what [847,293] does. The problem is architectural, not intellectual.

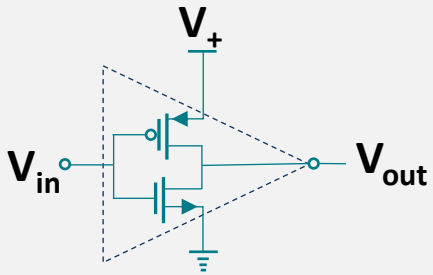
Simplified NN:



Electrical Engineering

An engineer can't read the current through transistor 31,847,293,004 on a live die — the silicon is sealed.

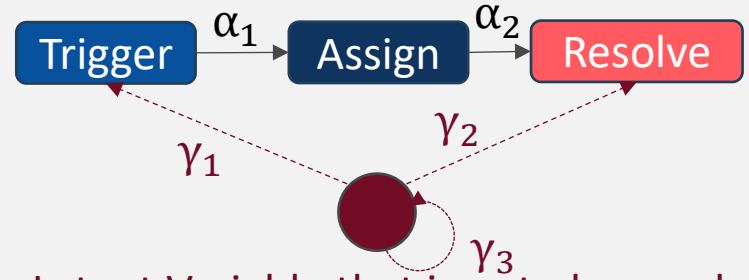
Simplified IC:



Business Process

No ops team can tell which of 2,847 steps caused today's SLA breach — the variable wasn't observed until the harm was done.

Simplified Business Process:



What do all these systems share? They all contain internal states that cannot be directly observed. We don't need them all — you only need to detect whether the critical path was followed.

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TAKEAWAY 1 · THE HINGE

The referee is the invariant.

An invariant is a quantity that stays fixed no matter how you measure it.

In a circuit, charge is conserved. It doesn't matter whether you use mesh or nodal analysis, label currents left or right — every valid method lands on the same number. When two engineers disagree, the conservation law says one of them is wrong and needs to check their work. The answer is discovered, **not chosen**— because charge and energy are conserved no matter which method, which components, or which sign convention you pick.

Attribution has no referee.

"Credit" isn't conserved. It doesn't exist until you pick a rule to invent it. Change the rule, change the answer — **and nothing in the world objects.**

Discovered vs. invented. That is the whole difference.

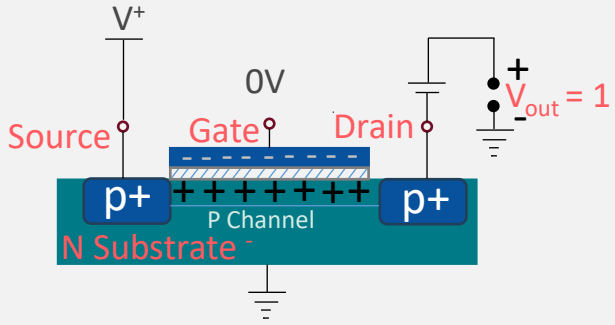
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ABSTRACTION IS NOT APPROXIMATION

The same silicon, described at three altitudes. **Each rung discards detail the rung above can't use — and choosing the altitude that matches the question is the whole skill.**

1 Carrier level

holes & electrons in silicon

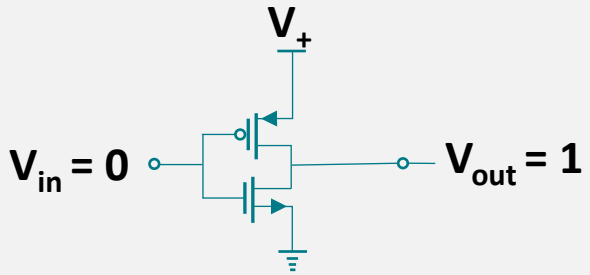


$$I_D = \frac{1}{2}\mu C_{OX}\frac{W}{L}(V_{GS} - V_{TH})^2$$

MODEL: device physics — hard but exact.
0 in → 1 out

2 Device level

FET as a switch



Low input in pulls output high

MODEL: 0 in → 1 out — the hinge

3 Logic level

Boolean function

A	Y
0	1
1	0

everything below has vanished

MODEL: Y = NOT A — pure function

1=1=1 no matter what level of abstraction

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ABSTRACTION IS NOT APPROXIMATION

The same conversion, described at three altitudes. **Here the totals match but the per-channel numbers don't survive — the abstraction discards the very thing the question needs.**

1 Journey level

the ordered, timed path

Display ↓ *day 0*

Search ↓ *+9 days*

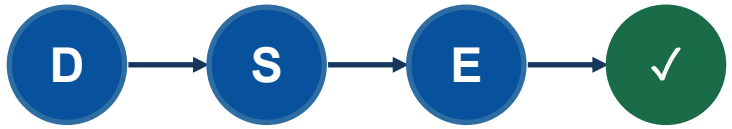
Email ↓ *+2 days*

CONVERT

MODEL: full DAG — order + lag. Hard but exact.

2 Path level

sequence as the hinge



display → search → email → convert

"credit depends on what PRECEDED what"

MODEL: sequence / removal effect — the hinge

3 Channel-set level

unordered bag + split

{ display, search, email }

Channel	Credit
display	30%
search	30%
email	40%

order has vanished

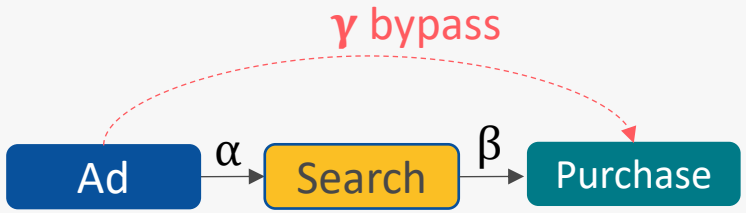
MODEL: Y = f(set) — invented split

Total conversions = same. Per-channel credit ≠ same. Level of abstraction changes the response. 1=1 ≠ 1

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THE SETUP · ONE MODEL

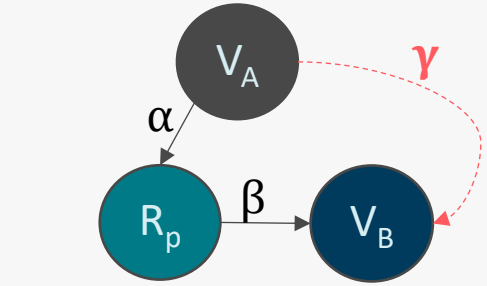
Business Process:
Declared Truth



γ : does the ad skip search and drive the buy directly?

=

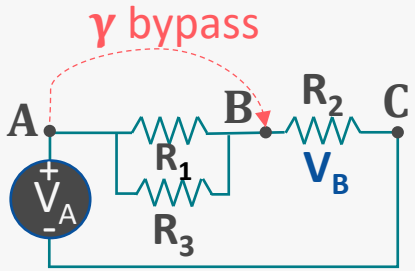
Structural Equation Model:
Declared Truth



→ Declared path (α, β)
⇢ Hidden bypass (γ)

=

Circuit Tested:
Declared Truth



$R_p = (R_1 \times R_3)/(R_1 + R_3)$
$V_B = V_A \times [(R_2)/(R_p + R_2)]$

γ : does Claude R_p computation?

We want to know:

- Does Gen AI skip a step in the declared truth?
- How do we know we prompted Gen AI correctly?
- How do we know it isn't a combination of prompting and skipping?

Why start with a circuit?

Three domains, one model. We prove the method where the answer is known — the circuit — then carry it to marketing, where it isn't.

1 Engineering climbed past aggregation.

A circuit obeys physical laws — what flows in must flow out. The true answer is fixed, so you can run the test and check the result against a known truth.

2 So, the circuit is the control condition.

It's where the method can be calibrated and trusted — you can prove it flags a real bypass, and a clean result means something. It's also where the **physics** caught **my own measurement** error.

3 Marketing is climbing the same mountain.

Marketing has mastered aggregation and is reaching for the same kind of fixed laws. It isn't there yet — not because the field lacks rigor, but because human behavior doesn't hand you a conserved quantity the way physics does. The truth isn't waiting in the data; it has to be built.

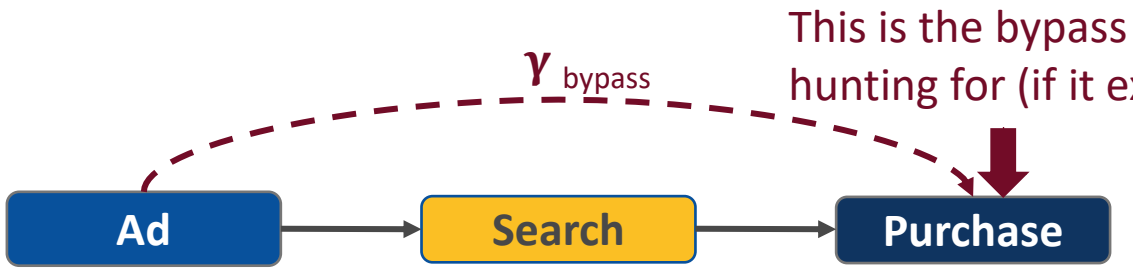
Prove the method on the circuit, where the answer is known. Then ask the same question of marketing — where it isn't.

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How do you catch a skipped step?

Forget the equations. The whole idea is one question: for any two steps, are they talking to each other directly — or only through the step in between?

The work passes through three tasks



Through the chain (good): Ad reaches Purchase only by going through Search. No direct line.

Back-channel (the alarm): Ad talks straight to Purchase — Search got skipped.

The readout: one cell per pair

	Ad	Search	Purchase
Ad	*	✓	= 0?
Search	✓	*	✓
Purchase	= 0?	✓	*

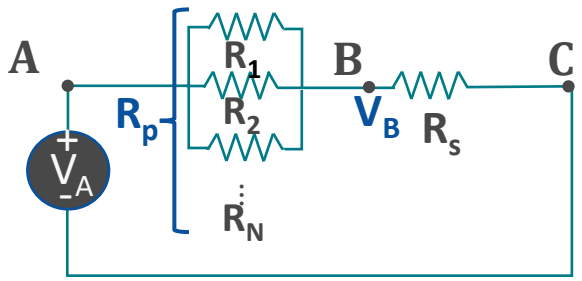
- ✓ **Neighbors connect — that's expected.**
- **Ad and Purchase should NOT connect directly.**

For the technically inclined: *the matrix above is the precision matrix, $Q = \Sigma^{-1}$ (inverse of the covariance matrix). A zero off-diagonal entry means two variables are conditionally independent given the rest — no direct path.*

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Here is what the **declared** truth looks like from a circuit and matrix point of view.

Declared Circuit



The schematic shows that A and B are wired only *through* R_p. No direct A-to-B link.

Should connect: V_A–R_p, R_p–V_B, V_B–R_S, R_p–R_S.

Should NOT connect: V_A–V_B, R_S–V_A.
If the mapping is honest, the forbidden cells (with red outlines) read = 0.

Precision Matrix

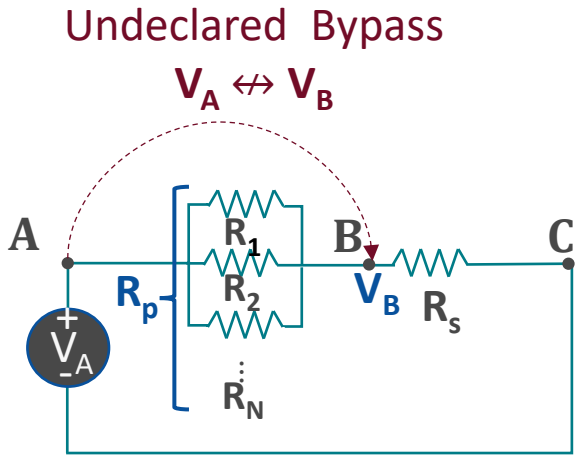
	V_A	R_p	R_S	V_B
V_A	*	$\neq 0$	$= 0$	$= 0$
R_p	$\neq 0$	*	$\neq 0$	$\neq 0$
R_S	$= 0$	$\neq 0$	*	$\neq 0$
V_B	$= 0$	$\neq 0$	$\neq 0$	*

If Gen AI is mapped correctly, the precision matrix will match the schematic

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Same circuit — but now with an undeclared shortcut. Here's what that does to the matrix.

Declared Circuit



Promised: A and B are wired only through R_p. No direct A-to-B link.

Actual: a hidden wire from A to B (red).

The forbidden (red outlined) cell flips from = 0 to ≠ 0. You can't hide a shortcut from this test.

Precision Matrix

	V_A	R_p	R_S	V_B
V_A	*	$\neq 0$	$= 0$	$\neq 0$
R_p	$\neq 0$	*	$\neq 0$	$\neq 0$
R_S	$= 0$	$\neq 0$	*	$\neq 0$
V_B	$\neq 0$	$\neq 0$	$\neq 0$	*

If Gen AI takes a shortcut, the precision matrix gives it away

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THE INSTRUMENT WAS RIGHT IT CAUGHT MY ERROR

Test:

The precision matrix checks more than the final answer. It tests whether each step was actually computed ***in order*** — was Step 2 really computed on the way to Step 4, or skipped?

1. What I first measured

	S1	S2	S3	S4
S1	*	$\neq 0$	$= 0$	$\neq 0$
S2	$\neq 0$	*	$= 0$	$\neq 0$
S3	$= 0$	$\neq 0$	*	$\neq 0$
S4	$\neq 0$	$\neq 0$	$\neq 0$	*

Step 1 ↔ Step 4 is not 0 — looks like a bypass. I believed it.

fix the parser

2. After fixing the instrument: the corrected result = hypothesis

	S1	S2	S3	S4
S1	*	$\neq 0$	$= 0$	$= 0$
S2	$\neq 0$	*	$= 0$	$\neq 0$
S3	$= 0$	$\neq 0$	*	$\neq 0$
S4	$= 0$	$\neq 0$	$\neq 0$	*

Step 1 ↔ Step 4 = 0 — no bypass. Step 2 was computed. Order followed.

3. The matrix was right

It flagged a real inconsistency. The fault was my parser reading the wrong number, not Claude skipping a step.

~25% → **~100%** on the hardest cases once the instrument read the model correctly.

A confident number needs something to check it against. In a circuit, that check is the physics. Gen AI and the engineer are strongest in concert — anchored to a truth neither one invents.

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THE PUZZLE

Two honest analysts, the same campaign data, contradictory conclusions about what drove the sale.

The method transfers from circuit to marketing. The referee — the invariant — does not.

First-touch says…

The awareness ad gets all the credit.
Conclusion: fund the top of the funnel.

Last-touch says…

The retargeting ad gets all the credit.
Conclusion: fund the bottom of the funnel.

Same customers. Same sales. Opposite budget decisions — and no test in the data settles which is right.

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THE SHARED WISDOM · WHY BOTH AGGREGATE

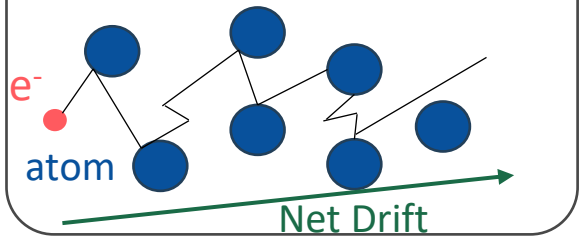
Engineers don't chase the electrons.

Engineering never predicts a circuit by tracking individual electrons — they drift, scatter, and bounce like a pinball. It works at the **aggregate** level: Ohm's law, Kirchhoff's laws. Lawful behavior that **emerges** from the chaos underneath

.

E **In a Wire** → Field points this way

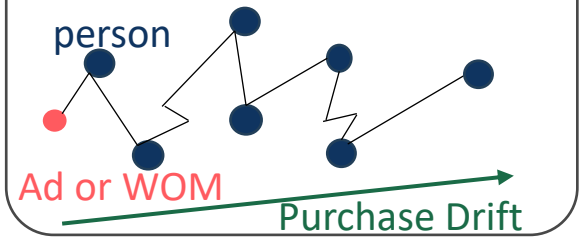
Electron in a wire



- **Random:** electron scatters between vibrating atoms
- **Directed:** Field tilts every deflection
- **Result:** Slow net drift, despite chaos

M **In a Market** → Motivation points this way

People in a Society



- **Random:** everyday churn, changing minds
- **Directed:** a shared nudge biases each choice
- **Result:** a collective shift, carried by WOM or Ad

Individual paths look random- the average has a direction

Marketing aggregates too — cohorts, funnels, touchpoints.

That is the correct instinct — the same one engineering learned.

Both fields work at the aggregate level. So why does one converge on a single answer and the other doesn't?

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TAKEAWAY 2 · THE DIAGNOSIS

Knobs, not measurements.

Attribution isn't one arbitrary choice. It's a stack of dials, none pinned to anything:

The model

First / Last / Sequential/Linear / U-shaped / Time-Decay

The window

40 days / 15 days/ 7 days — a cutoff influence never actually has

The decay

How fast credit fades, if weighted at all

In all of my marketing interviews, and speaking with people in industry, the 40-vs-15-day debate never ends — and that is the proof. If there were a right answer, the argument would have stopped.

THE LADDER · WHAT PINS THE ANSWER DOWN

Trust a number by what it's pinned to.

Conserved law	the circuit (charge)	Invariant by nature	**Self-enforcing**
Experiment	randomized incrementality	Unconditional invariant	**Falsifiable**
Quasi-experiment	DiD, synthetic control, IV, RDD	Conditional invariant	**Partly falsifiable**
Convention	first / last / multi-touch	No invariant	**Unfalsifiable**

Marketing can climb to experiment — but the top rung belongs to physics. Its invariant must be built, not found. Gen AI's reasoning can be verified where there's an invariant (top), and can only be taken on faith where there isn't (bottom)

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TAKEAWAY 3 · WHERE THIS IS GOING

Marketing is halfway up the mountain.

Engineering reached emergence — stable, predictive laws — by first discovering what is conserved.

Marketing has mastered aggregation. It hasn't found its conservation laws yet. With human behavior and an ever-changing environment, that's a hard climb. But it isn't a deficient science — it's a climbing one. The substrate is human, so the invariant must be built, not found.

The way up

- Structure first — formulate a model, find the leaks.
- Test it with an experiment that supplies a real invariant.
- Gen AI is the starting instrument, not the conservation law.

Not better conventions. Not an AI oracle. Invariants.

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A number you can't prove wrong isn't a measurement.

It's a choice wearing a costume.

Marketing creates real value.

But we don’t yet have the science to cleanly connect spend, targeting, and ROI — and no experiment-free attribution window can be proven more correct than another.

The danger isn’t marketing; it’s a culture that rewards a confident number over an honest *“we can’t know that yet.”*

Pretending to a precision we don’t have isn’t rigor. It’s what eventually breaks trust in the whole function.

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